Exercises

Logic Programming

https://moodle.aau.at/course/view.php?id=47556

University of Klagenfurt — Winter semester 2026

Exercise 1 (Demonstration: 21 October 2026)

Quiz

		Answer	
1.	Each positive logic program has at most one model.	**true**	**false**
2.	Each positive logic program has at most one answer set.	**true**	**false**
3.	Each normal logic program has at least one model.	**true**	**false**
4.	Each normal logic program has at least one answer set.	**true**	**false**

Exercise 1.1

Determine the answer set of the following positive logic programs:

1.1–a

```
a.
b :- e.
b :- f.
c :- b, f.
c :- d, g.
d.
e :- b, d.
g :- a, b.
g :- a, d.
```

1.1–b

```
likes(kate,fish).
likes(kate,soup).
likes(mike,beef).
likes(mike,cake).
likes(paul,duck).
likes(paul,pork).
likes(tina,beef).
likes(tina,soup).

% order(Z) :- likes(X,Z), likes(Y,Z), X != Y.
order(beef) :- likes(mike,beef), likes(tina,beef).
order(soup) :- likes(kate,soup), likes(tina,soup).

% happy(X) :- likes(X,Y), order(Y).
happy(kate) :- likes(kate,fish), order(fish).
happy(kate) :- likes(kate,soup), order(soup).
happy(mike) :- likes(mike,beef), order(beef).
happy(mike) :- likes(mike,cake), order(cake).
happy(paul) :- likes(paul,duck), order(duck).
happy(paul) :- likes(paul,pork), order(pork).
happy(tina) :- likes(tina,beef), order(beef).
happy(tina) :- likes(tina,soup), order(soup).
```

Exercise 1.2

Consider the following sets of (true) atoms:

- $X_1 = \{\texttt{a}, \texttt{b}, \texttt{d}, \texttt{e}\}$
- $X_2 = \{\texttt{b}, \texttt{e}\}$

Decide which of the above sets of atoms are answer sets for the following normal logic programs:

1.2–a

```
a :- b.
b :- not c.
c :- not b.
d :- a, e.
e.
```

1.2–b

```
a :- not b.
a :- d.
b :- not c.
c :- a, not b.
d :- a.
d :- not c, not e.
e :- b, not a.
e :- d.
```

Exercise 1.3

Reconsider the two logic programs in the previous exercise and find all their answer sets. (HINT: Avoid considering all of the possible interpretations.)

Exercise 1.4

Determine the answer sets of the following normal logic programs:

1.4–a

```
canary(tweety).
penguin(tux).

bird(tweety) :- canary(tweety).
bird(tux) :- penguin(tux).

injured(tweety) :- not ok(tweety), bird(tweety).
ok(tweety) :- not injured(tweety), bird(tweety).
injured(tux) :- not ok(tux), bird(tux).
ok(tux) :- not injured(tux), bird(tux).

flies(tweety) :- ok(tweety), not penguin(tweety).
flies(tux) :- ok(tux), not penguin(tux).
```

1.4–b

```
num(1). num(2). num(3).

lucky(1) :- num(1), not unlucky(1).
unlucky(1) :- num(1), not lucky(1).
lucky(2) :- num(2), not unlucky(2).
unlucky(2) :- num(2), not lucky(2).
lucky(3) :- num(3), not unlucky(3).
unlucky(3) :- num(3), not lucky(3).

p :- unlucky(1), not p.
p :- unlucky(2), not p.
p :- unlucky(3), not p.
```